

1.2 The notion of limit

Motivating Questions

- What is the mathematical notion of *limit* and what role do limits play in the study of functions?
 - What is the meaning of the notation $\lim_{x \rightarrow a} f(x) = L$?
 - How do we go about determining the value of the limit of a function at a point?
 - How do we manipulate average velocity to compute instantaneous velocity?
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1.2.1 Introduction

In Section 1.1 we used a function, $s(t)$, to model the location of a moving object at a given time. Functions can model other interesting phenomena, such as the rate at which an automobile consumes gasoline at a given velocity, or the reaction of a patient to a given dosage of a drug. We can use calculus to study how a function value changes in response to changes in the input variable.

Think about the falling ball whose position function is given by $s(t) = 64 - 16t^2$. Its average velocity on the interval $[1, x]$ is given by

$$AV_{[1,x]} = \frac{s(x) - s(1)}{x - 1} = \frac{(64 - 16x^2) - (64 - 16)}{x - 1} = \frac{16 - 16x^2}{x - 1}.$$

Note that the average velocity is a function of x . That is, the function $g(x) = \frac{16 - 16x^2}{x - 1}$ tells us the average velocity of the ball on the interval from $t = 1$ to $t = x$. To find the instantaneous velocity of the ball when $t = 1$, we need to know what happens to $g(x)$ as x gets closer and closer to 1. But also notice that $g(1)$ is not defined, because it leads to the quotient $0/0$.

This is where the notion of a *limit* comes in. By using a limit, we can investigate the behavior of $g(x)$ as x gets arbitrarily close, but not equal, to 1. We first use the graph of a function to explore points where interesting behavior occurs.

Preview Activity 1.2.1. Suppose that g is the function given by the graph below. Use the graph in Figure 1.2.1 to answer each of the following questions.

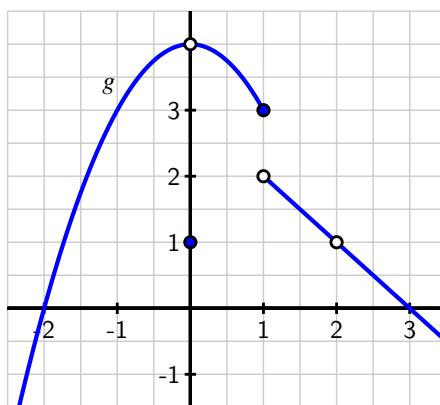


Figure 1.2.1 Graph of $y = g(x)$ for Preview Activity 1.2.1.

- Determine the values $g(-2)$, $g(-1)$, $g(0)$, $g(1)$, and $g(2)$, if defined. If the function value is not defined, explain what feature of the graph tells you this.
- For each of the values $a = -1$, $a = 0$, and $a = 2$, complete the following sentence: "As x gets closer and closer (but not equal) to a , $g(x)$ gets as close as we want to _____."
- What happens as x gets closer and closer (but not equal) to $a = 1$? Does the function $g(x)$ get as close as we would like to a single value?

1.2.2 The Notion of Limit

Limits give us a way to identify a trend in the values of a function as its input variable approaches a particular value of interest. We need a precise understanding of what it means to say "a function f has limit L as x approaches a ." To begin, think about a recent example.

In Preview Activity 1.2.1, we saw that as x gets closer and closer (but not equal) to 0, $g(x)$ gets as close as we want to the value 4. At first, this may feel counterintuitive, because the value of $g(0)$ is 1, not 4. But limits describe the behavior of a function *arbitrarily close to* a fixed input, and the value of the function *at* the fixed input does not matter. More formally,¹ we say the following.

Definition 1.2.2 Given a function f , a fixed input $x = a$, and a real number L , we say that f **has limit L as x approaches a** , and write

$$\lim_{x \rightarrow a} f(x) = L$$

provided that we can make $f(x)$ as close to L as we like by taking x sufficiently close (but not

¹What follows here is not what mathematicians consider the formal definition of a limit. To be completely precise, it is necessary to quantify both what it means to say "as close to L as we like" and "sufficiently close to a ." That can be accomplished through what is traditionally called the epsilon-delta definition of limits. The definition presented here is sufficient for the purposes of this text.

equal) to a . If we cannot make $f(x)$ as close to a single value as we would like as x approaches a , then we say that f **does not have a limit as x approaches a** . $\diamond$

Example 1.2.3 For the function g pictured in Figure 1.2.1 from Preview Activity 1.2.1, what can we say about the values of

$$\lim_{x \rightarrow -1} g(x), \lim_{x \rightarrow 0} g(x), \lim_{x \rightarrow 1} g(x), \text{ and } \lim_{x \rightarrow 2} g(x)?$$

Solution. When working from a graph, it suffices to ask if the function *approaches* a single value from each side of the fixed input. The function value *at* the fixed input is irrelevant. Based on the graph of g in Figure 1.2.1, we can conclude that

$$\lim_{x \rightarrow -1} g(x) = 3, \lim_{x \rightarrow 0} g(x) = 4, \text{ and } \lim_{x \rightarrow 2} g(x) = 1,$$

because at each of those x -values, the function approaches the same height from both sides of the point of interest.

However, g does not have a limit as $x \rightarrow 1$ due to the jump in the graph at $x = 1$. If we approach $x = 1$ from the left, the function values tend to get close to 3, but if we approach $x = 1$ from the right, the function values get close to 2. There is no single number that all of these function values approach. This is why the limit of g does not exist at $x = 1$. $\diamond$

For any function f , there are typically three ways to answer the question “does f have a limit at $x = a$, and if so, what is the limit?” The first is to reason graphically as we have just done with the example from Preview Activity 1.2.1. If we have a formula for $f(x)$, there are two additional possibilities:

1. Evaluate the function at a sequence of inputs that approach a on either side (typically using some sort of computing technology), and ask if the sequence of outputs seems to approach a single value.
2. Use the algebraic form of the function to understand the trend in its output values as the input values approach a .

The first approach produces only an approximation of the value of the limit, while the latter can often be used to determine the limit exactly.

Example 1.2.4 Limits of Two Functions. For each of the following functions, we’d like to know whether or not the function has a limit at the stated a -values. Use both numerical and algebraic approaches to investigate and, if possible, estimate or determine the value of the limit. Compare the results with a careful graph of the function on an interval containing the points of interest.

a. $f(x) = \frac{4-x^2}{x+2}; a = -1, a = -2$

b. $g(x) = \sin\left(\frac{\pi}{x}\right); a = 3, a = 0$

Solution. a. We first construct a graph of f along with tables of values near $a = -1$ and $a = -2$.

x	$f(x)$	x	$f(x)$
-0.9	2.9	-1.9	3.9
-0.99	2.99	-1.99	3.99
-0.999	2.999	-1.999	3.999
-0.9999	2.9999	-1.9999	3.9999
-1.1	3.1	-2.1	4.1
-1.01	3.01	-2.01	4.01
-1.001	3.001	-2.001	4.001
-1.0001	3.0001	-2.0001	4.0001

Table 1.2.5 Values of f near $x = -1$.

Table 1.2.6 Values of f near $x = -2$.

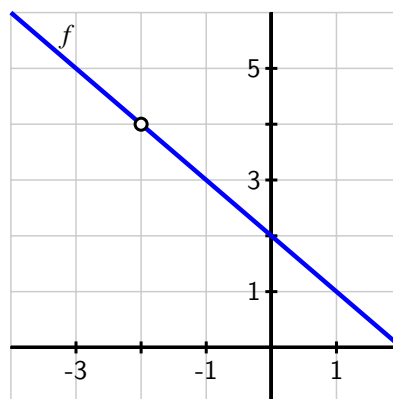


Figure 1.2.7 Plot of $f(x)$ on $[-4, 2]$.

From Table 1.2.5, it appears that we can make f as close as we want to 3 by taking x sufficiently close to -1 , which suggests that $\lim_{x \rightarrow -1} f(x) = 3$. This is also consistent with the graph of f . To see this a bit more rigorously and from an algebraic point of view, consider the formula for f : $f(x) = \frac{4-x^2}{x+2}$. As $x \rightarrow -1$, $(4 - x^2) \rightarrow (4 - (-1)^2) = 3$, and $(x + 2) \rightarrow (-1 + 2) = 1$, so as $x \rightarrow -1$, the numerator of f tends to 3 and the denominator tends to 1, hence $\lim_{x \rightarrow -1} f(x) = \frac{3}{1} = 3$.

The situation is more complicated when $x \rightarrow -2$, because $f(-2)$ is not defined. If we try to use a similar algebraic argument regarding the numerator and denominator, we observe that as $x \rightarrow -2$, $(4 - x^2) \rightarrow (4 - (-2)^2) = 0$, and $(x + 2) \rightarrow (-2 + 2) = 0$, so as $x \rightarrow -2$, the numerator and denominator of f both tend to 0. We call $0/0$ an *indeterminate form*. This tells us that there is more work to do to try to determine the limit's value. From Table 1.2.6 and Figure 1.2.7, it appears that f should have a limit of 4 at $x = -2$.

To see algebraically why this is the case, observe that

$$\begin{aligned} \lim_{x \rightarrow -2} f(x) &= \lim_{x \rightarrow -2} \frac{4 - x^2}{x + 2} \\ &= \lim_{x \rightarrow -2} \frac{(2 - x)(2 + x)}{x + 2}. \end{aligned}$$

It is important to observe that, since we are taking the limit as $x \rightarrow -2$, we are considering x values that are close, but not equal, to -2 . Because we never actually allow x to equal -2 , the quotient $\frac{2+x}{x+2}$ has value 1 for every possible value of x . Thus, we can simplify the most recent expression above, and find that

$$\lim_{x \rightarrow -2} f(x) = \lim_{x \rightarrow -2} 2 - x.$$

This limit is now easy to determine, and its value clearly is 4. Thus, from several points of view we've seen that $\lim_{x \rightarrow -2} f(x) = 4$.

b. Next we turn to the function g , and construct two tables and a graph.